

SATYAM KUMAR

AI SYSTEMS ARCHITECT &
BACKEND DEVELOPER

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EDUCATION

INDIAN INSTITUTE OF INFORMATION
TECHNOLOGY, KALYANI

B.Tech in Computer Science and Engineering
Expected June 2021 | CGPA - 9.26/10

SKILLS

LLM orchestration, Prompt optimization, Vector search, RAG pipelines, API governance, Model evaluation, Python, JavaScript, Typescript, PostgreSQL, MongoDB, Node.js, Redis, RabbitMQ, Kafka

OPEN SOURCE

TATA 1MG

- Built DeputyDev, an AI-powered VS Code assistant and CLI for GenAI code generation, spanning 5 repositories with major contributions across backend, analytics, and editor tooling, boosting developer productivity and feature delivery.

KIWIX | CONTRIBUTIONS | RECOMMENDATION

PYTHON DEVELOPER (FREELANCE) | AUG 2020 - DEC 2020

- Designed a better version of OpenEdx scraper, a single tool to create a ZIM out of any OpenEdx powered system (Eg. Edx).
- Worked on libraries for various stages in ZIM creation

STUDENT DEVELOPER DURING GOOGLE SUMMER OF CODE | JUN 2020 - AUG 2020

- Improved Python offliners and Zimfarm to archive web content into ZIM files, achieving ~10× faster offlining and up to 50% smaller ZIMs.
- Made major contributions to seven projects; selected among 1,199 participants in Google Summer of Code.

EXPERIENCE

TATA 1mg | APP

SDE II | APRIL - 2024 - PRESENT

- Led cross-functional adoption of **GenAI tools** across 1mg's developer ecosystem, improving **engineering throughput by 30%+**.
- Built and led **DeputyDev**, an AI-powered VS Code assistant and CLI for GenAI code generation using **multiple LLMs**, **boosting developer productivity and accelerating delivery** across 1mg.
- Designed a **typed, stateless multi-model backend** with region-aware rate limiting, prompt caching, and adaptive chunking, **cutting latency/errors 60%+** and lowering costs.
- Overhauled query-solving, history, and intent systems, **improving speed 3–4×**, handling **1,000+ daily queries** reliably, and **reducing chat errors 300+ → <100/day**.
- Developed a **custom VS Code diff applicator** enabling seamless AI-driven code edits in "Act" mode.
- Built **real-time analytics dashboards** and refactored backend (Python → Astral + UV + Ruff), **cutting setup time hours → minutes** and improving uptime.
- Mentored engineers and launched **Lab Buddy (Diagnostics)** using Redis vector search, **accelerating internal tool delivery**.
- **Re-architected payments/refunds**, reducing memory 50% and latency 30%, enabling faster order processing.
- Enhanced **Care Plan and Loyalty systems**, **6× adoption growth (200 → 1,200/day)**, boosting retention/revenue and **cutting integration time 30×**.

SDE I | OCTOBER - 2022 - APRIL - 2024

- **Enhanced loyalty/rewards service** for fault tolerance.
- **Created a system to onboard third-party partnerships** via simple APIs, saving **30× implementation time**.

YELLOW CLASS (now SpeakX) | WEBSITE

SOFTWARE DEVELOPMENT ENGINEER | JULY - 2021 - OCT - 2022

- Built **microservices** to improve performance and scale for a top 10 Indian education app.
- Developed **communications backend**, enabling 5× growth in user interactions.
- Created **Quiz module** from scratch, handling 1,000+ req/min with minimal resource use.
- Designed **URL shortener** with caching and message queues for low-latency performance.
- Maintained libraries, introduced **strict typing**, and built a reusable **nested discriminator library** in Mongoose.
- Implemented backend for **mini-games, collectibles, and rewards**, boosting engagement and retention.

AWARDS

2025	Organizational	Received Department Award - Performance for contributions to DeputyDev and other GenAI based projects
2020	National	Winner of Smart India Hackathon 2020 Software under Problem Statement DR135 by Govt. of Goa
2019	National	"2nd Runner-up" in CodeUtsava 3.0 Hackathon at NIT Raipur
2018	National	"Most Innovative Hardware Hack" in HACKABIT 2018
2017	National	2nd rank in SOF National Cyber Olympiad (NCO) in India
2016	Zonal	2nd rank in finals of SOF National Cyber Olympiad in Jharkhand